

ANEJ SVETE

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EDUCATION

- ETH Zürich, ETH AI Center**, *PhD in Computer Science* 2023–Present
- Advisors: Ryan Cotterell and Valentina Boeva
 - Research interests: Analysis of language models with formal languages, interpretability of representation learning
- ETH Zürich**, *Master of Data Science* 2020–2023
- GPA: 5.84/6; highest in the Master's program cohort.
 - Relevant Courses: Deep Learning, Advanced Machine Learning, Optimization for Data Science, Natural Language Processing, Fundamentals of Mathematical Statistics, Statistical Modeling, Causality
- University of Ljubljana**, *Bachelor of Computer Science and Mathematics* 2017–2020
- GPA: 10/10; top student in the department each year.
- Stanford University**, *two courses as part of my bachelor's degree* 2019–2020
- Technical Gymnasium Ljubljana**, *Gymnasium* 2013–2017
- Perfect score on Matura examination: 34/34

WORK AND TEACHING EXPERIENCE

- Research Assistant**, *Allen Institute for AI* 2025
- A research investigating the efficiency and reasoning abilities of diffusion language models.
- Head Teaching Assistant**, *ETH Zürich*
- Teaching (600+ students), material preparation, course organization, and managing a team of 20+ teaching assistants.
- Large Language Models** 2022–Present
- Natural Language Processing** 2022–2023
- Advanced Formal Language Theory** 2021–2023
- Teaching Assistant**, *ETH Zürich*
- Philosophy of Language and Computation** 2022–2023
- Machine Learning for Healthcare** 2022
- Machine Learning for Genomics** 2022
- Computational Biomedicine** 2021–2022
- Software Developer**, *XLAB* 2017–2020
- Tutor for Mathematics**, *University of Ljubljana* 2018–2019

BIBLIOGRAPHY

[Training neural networks as recognizers of formal languages](#) (ICLR 2025).

Alexandra Butoi, Ghazal Khalighinejad, **Anej Svete**, Josef Valvoda, Ryan Cotterell, and Brian DuSell.

[Unique hard attention: A tale of two sides](#) (ACL 2025).

Selim Jerad, **Anej Svete**, Jiaoda Li, and Ryan Cotterell.

[Gumbel counterfactual generation from language models](#) (ICLR 2025).

Shauli Ravfogel^{*}, **Anej Svete**^{*}, Vésteinn Snæbjarnarson, and Ryan Cotterell.

[What languages are easy to language-model? A perspective from learning probabilistic regular languages](#) (ACL 2024).

Nadav Borenstein, **Anej Svete**, Robin Chan, Josef Valvoda, Franz Nowak, Isabelle Augenstein, Eleanor Chodroff, and Ryan Cotterell.

[Computational expressivity of neural language models](#) (ACL 2024).

Alexandra Butoi, Robin Chan, Ryan Cotterell, William Merrill, Franz Nowak, Clemente Pasti, Lena Strobl, and **Anej Svete**.

[On affine homotopy between language encoders](#) (NeurIPS 2024).

Robin SM Chan, Reda Boumasmoud, **Anej Svete**, Yuxin Ren, Qipeng Guo, Zhijing Jin, Shauli Ravfogel, Mrinmaya Sachan, Bernhard Schölkopf, Mennatallah El-Assady, and Ryan Cotterell.

[On the representational capacity of neural language models with chain-of-thought reasoning](#) (ACL 2024).

Franz Nowak, **Anej Svete**, Alexandra Butoi, and Ryan Cotterell.

[An \$L^*\$ algorithm for deterministic weighted regular languages](#) (EMNLP 2024).

Clemente Pasti, Talu Karagöz, Franz Nowak, **Anej Svete**, Reda Boumasmoud, and Ryan Cotterell.

[A probability–quality trade-off in aligned language models and its relation to sampling adaptors](#) (EMNLP 2024).

Naaman Tan, Josef Valvoda, Tianyu Liu, **Anej Svete**, Yanxia Qin, Min-Yen Kan, and Ryan Cotterell.

[Can transformers learn \$n\$ -gram language models?](#) (EMNLP 2024).

Anej Svete, Nadav Borenstein, Mike Zhou, Isabelle Augenstein, and Ryan Cotterell.

[On efficiently representing regular languages as RNNs](#) (ACL 2024).

Anej Svete, Robin Chan, and Ryan Cotterell.

[Transformers can represent \$n\$ -gram language models](#) (NAACL 2024).

Anej Svete and Ryan Cotterell.

[Lower bounds on the expressivity of recurrent neural language models](#) (NAACL 2024).

Anej Svete, Franz Nowak, Anisha Sahabdeen, and Ryan Cotterell.

[Formal aspects of language modeling.](#)

Ryan Cotterell, **Anej Svete**, Clara Meister, Tianyu Liu, and Li Du.

[A geometric notion of causal probing](#) (arXiv preprint 2023).

Clément Guerner, **Anej Svete**, Tianyu Liu, Alexander Warstadt, and Ryan Cotterell.

[On the representational capacity of recurrent neural language models](#) (EMNLP 2023).

Franz Nowak, **Anej Svete**, Li Du, and Ryan Cotterell.

[Efficiently representing finite-state automata with recurrent neural networks](#) (arXiv preprint 2023).

Anej Svete and Ryan Cotterell.

[Recurrent neural language models as probabilistic finite-state automata](#) (EMNLP 2023).

Anej Svete and Ryan Cotterell.

[Algorithms for acyclic weighted finite-state automata with failure arcs](#) (EMNLP 2022).

Anej Svete, Benjamin Dayan, Ryan Cotterell, Tim Vieira, and Jason Eisner.

Ne gre le za melodijo: Kako Evropa glasuje za svoje najljubše skladbe (Uporabna informatika 2020).

Anej Svete, Jakob Hostnik, and Lovro Šubelj.

SCHOLARSHIPS AND AWARDS

ETH AI Center Scholarship , a graduate scholarship for exceptional PhD candidates in AI at ETH Zürich.	2023–Present
Willi Studer Prize for the highest GPA in the study program at the end of my master's studies.	2023
Ad Futura Scholarship , awarded to excellent students studying abroad by the Slovenian government.	2020–2022
Rector's Prize for exceptional study results at the end of my bachelor's studies.	July 2021
Dean's Prize for exceptional study results all three years of my bachelor's studies.	2018–2020
Intern Scholarship by XLAB during my bachelor's studies.	2017–2020
Golden Owl Award , awarded by the president of the Republic of Slovenia to students who achieve all possible points on the national matriculation examination. I was one of 16 awardees that year.	2017

SERVICE

Organizer of the Seminar on Formal Languages and Neural Networks (FLaNN)	2025
ICML Tutorial on Language Models and Formal Language Theory	2025
ACL Tutorial on Language Models and Formal Language Theory (Leader)	2024
ESSLLI Course on Language Models and Formal Language Theory (Lecturer)	2023
NLP Tutorial at the International Committee of the Red Cross (Organizer)	2023
Reviewer: ICLR, ACL, EMNLP, NAACL	2023–Present

SKILLS

Languages: Slovene (native), English (fluent), German (upper-intermediate)

Programming Languages : Python, Python Data Science tools (numpy, pytorch, pytorch lightning, pandas, scikit-learn, huggingface), R, Java, Matlab, SQL

Digital: Unix, $\text{\LaTeX}$